PROJECT WORK

This project was done by members of **GROUP 11.** Below is a table of each student with his or her respective index number.

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1. ABSTRACT

The sheets below considers the modelling of a life table for Ghana. To attain this, we select the data from the year 2015 of our life table. Life tables are computed from age and the survival curves compared with other survival curves to see the differences. A life table is estimated by fitting the survival curve of the generated life table to already existing mortality models. The mortality model is said to follow a De Moivre Distribution with a survival function $S_x = \left(\frac{\omega - x}{\omega}\right)^{\alpha}$ $0 \leq x \leq \omega \; and \; \alpha > 0$.

$\omega = 90$ And i=6%

2. INTRODUCTION

- **Background**

A life table (also called a mortality table or actuarial table) is a table which shows, for each age, what the probability is that a person of that age will die before their next birthday ("probability of death "). In other words, it represents the survivorship of people from a certain population. They can also be explained as a long-term mathematical way to measure a population's longevity. Life (mortality) tables are very useful demographic tools. A basic life table is just a list of the number of individuals alive at each age and the number of deaths in each period.

- **Life Tables and Actuarial Values for Ghana**

There are few number of life tables available for Ghana. It includes life tables from Ghana Statistical Service and World Health Organization. These tables were computed from population census with some little modifications and propositions made.

3. DESCRIPTION OF A LIFE TABLE

A typical lifetable looks like the table now (where n =1).

Age(x)	l_x	${}_nd_x$	${}_nq_x$	${}_np_x$	S_x	e_x	A_x	$\ddot{a}$	P_x
0	10000								
1									
2									
⋮									
100									

l_X = The number of people alive at age x.

${}_nd_x$= number of deaths at age x. $ndx = l_x - l_{x+n}$

${}_nq_x$= probability that a person aged x dies before n years. $nqx = 1 - npx$.

${}_np_x$= probability of surviving from age x to x + n. $npx = \frac{l_{x+n}}{l_x}$

e_x = the expectation of life at age x. $e_x = \frac{T_x}{l_x}$

A_x= actuarial present values for a discrete whole life insurance policy.

$Ax = \sum_{k=o}^{w-x-90} V^{k+5} * {}_{k|5}q_x$, where k=0, 5, 10,…, ϖ-5.

$\ddot{a}$= actuarial present value for a discrete whole life annuity policy. $\ddot{a} = \frac{1-A_x}{d}$

P_x = the whole life benefit insurance premium for a life age x. $P_x = \frac{A_x}{(\ddot{a}_x)}$

Below is a statistical data provided for Ghana by the World Health Organisation.

		2019	2019	2019	2015	2015	2015	2010	2010	2010	2005	2005	2005	2000	2000	2000
Indicator	Age Group	Both sexe	Male	Female	Both sexe	Male	Female	Both sexe	Male	Female	Both sexe	Male	Female	Both sexe	Male	Female
nMx - age	<1 year	0.035108	0.039004	0.03104	0.039425	0.043673	0.034979	0.049019	0.054436	0.043379	0.058445	0.064559	0.052044	0.066686	0.073198	0.05987
nMx - age	1-4 years	0.003165	0.003411	0.00291	0.00414	0.00442	0.003848	0.00593	0.006262	0.005584	0.007556	0.007898	0.0072	0.009568	0.009918	0.009202
nMx - age	5-9 years	0.001201	0.001294	0.001103	0.001381	0.001475	0.001284	0.001652	0.001745	0.001556	0.001996	0.002087	0.001902	0.002407	0.002495	0.002315
nMx - age	10-14 yea	0.000947	0.001103	0.000785	0.001027	0.001185	0.000862	0.001149	0.001316	0.000974	0.001294	0.001473	0.001106	0.001456	0.001624	0.00128
nMx - age	15-19 yea	0.001263	0.001553	0.00096	0.001444	0.001746	0.001129	0.001584	0.001868	0.001286	0.001594	0.001835	0.001341	0.001559	0.001701	0.001411
nMx - age	20-24 yea	0.001738	0.002149	0.001308	0.001968	0.002365	0.001554	0.002249	0.002555	0.00193	0.00243	0.002544	0.002312	0.002508	0.002407	0.002613
nMx - age	25-29 yea	0.002134	0.00242	0.001833	0.002498	0.002697	0.002291	0.003039	0.002971	0.00311	0.003575	0.0031	0.004062	0.003682	0.003046	0.004333
nMx - age	30-34 yea	0.002882	0.003133	0.002619	0.003444	0.003544	0.003341	0.00428	0.003986	0.004582	0.005074	0.004401	0.005761	0.005064	0.004409	0.005734
nMx - age	35-39 yea	0.004048	0.004469	0.003612	0.004798	0.005061	0.004527	0.005829	0.005748	0.005912	0.006799	0.006598	0.007004	0.006516	0.006439	0.006594
nMx - age	40-44 yea	0.005786	0.006699	0.004853	0.006704	0.007589	0.005803	0.007907	0.008617	0.007185	0.008834	0.009603	0.008057	0.008326	0.008994	0.00766
nMx - age	45-49 yea	0.007883	0.009499	0.006254	0.008909	0.010609	0.007194	0.010237	0.011881	0.008587	0.011074	0.012594	0.009577	0.010506	0.011492	0.009543
nMx - age	50-54 yea	0.011157	0.013876	0.00845	0.012369	0.015362	0.009404	0.013923	0.016961	0.010966	0.014758	0.017389	0.012226	0.01429	0.016074	0.012575
nMx - age	55-59 yea	0.015483	0.019566	0.011529	0.016891	0.021398	0.012588	0.01878	0.023451	0.014368	0.019494	0.023622	0.015616	0.01897	0.022044	0.016208
nMx - age	60-64 yea	0.023085	0.02922	0.0174	0.02491	0.031665	0.018691	0.027284	0.034419	0.02074	0.027889	0.034511	0.022102	0.027537	0.032596	0.023064
nMx - age	65-69 yea	0.033273	0.041647	0.025807	0.035556	0.044838	0.027284	0.038292	0.04834	0.029735	0.039279	0.048516	0.031344	0.039106	0.046292	0.032749
nMx - age	70-74 yea	0.050954	0.061937	0.041532	0.053694	0.066221	0.043359	0.057688	0.070869	0.046678	0.059333	0.071427	0.048951	0.059397	0.068715	0.051221
nMx - age	75-79 yea	0.075131	0.08883	0.06437	0.078724	0.094188	0.066306	0.084048	0.100161	0.070692	0.086076	0.101124	0.073341	0.086089	0.097955	0.076208
nMx - age	80-84 yea	0.116137	0.132836	0.103307	0.12085	0.140158	0.105671	0.128096	0.148231	0.111762	0.129836	0.149348	0.114301	0.129996	0.145728	0.117708
nMx - age	85+ years	0.195563	0.216696	0.1798	0.197182	0.221051	0.179297	0.203538	0.228206	0.185508	0.204861	0.228899	0.187649	0.20611	0.2269	0.191525
nqx - prob	<1 year	0.034266	0.037967	0.03038	0.038366	0.042378	0.034143	0.047393	0.052438	0.042101	0.056148	0.061768	0.050215	0.063712	0.06963	0.057461
nqx - prob	1-4 years	0.012566	0.013532	0.011558	0.016397	0.017493	0.015252	0.023387	0.024679	0.022042	0.029685	0.031003	0.028309	0.037412	0.038751	0.036012
nqx - prob	5-9 years	0.005985	0.006447	0.005502	0.006883	0.007347	0.0064	0.008226	0.008685	0.007748	0.009932	0.010379	0.009466	0.011962	0.012398	0.011508
nqx - prob	10-14 yea	0.004723	0.005497	0.003915	0.005122	0.005907	0.004302	0.005727	0.00656	0.004856	0.006447	0.007337	0.005516	0.007252	0.008088	0.006377
nqx - prob	15-19 yea	0.006296	0.007734	0.00479	0.007194	0.00869	0.005628	0.007888	0.009299	0.00641	0.007936	0.009133	0.006684	0.007764	0.008468	0.007032
nqx - prob	20-24 yea	0.008652	0.010688	0.006517	0.009794	0.011754	0.007738	0.011184	0.012694	0.009605	0.012078	0.01264	0.011496	0.012464	0.011962	0.012982
nqx - prob	25-29 yea	0.010612	0.012029	0.009126	0.012414	0.013396	0.011387	0.015081	0.014744	0.01543	0.017717	0.015383	0.020105	0.018242	0.015115	0.021435
nqx - prob	30-34 yea	0.014305	0.015544	0.013009	0.017075	0.017566	0.016566	0.021175	0.019734	0.022651	0.025054	0.021764	0.028397	0.025004	0.021804	0.028267
nqx - prob	35-39 yea	0.020035	0.0221	0.017896	0.023703	0.02499	0.022381	0.028728	0.028332	0.029131	0.033428	0.032455	0.034418	0.032056	0.031684	0.032434
nqx - prob	40-44 yea	0.028518	0.032945	0.023973	0.032967	0.03724	0.028601	0.038766	0.042175	0.035289	0.043214	0.04689	0.039489	0.040781	0.043983	0.03758
nqx - prob	45-49 yea	0.038656	0.046392	0.030787	0.043576	0.051673	0.035335	0.049907	0.057693	0.042033	0.053878	0.061046	0.046765	0.051187	0.055854	0.046602
nqx - prob	50-54 yea	0.054272	0.067056	0.041375	0.05999	0.073969	0.045942	0.067274	0.081355	0.053366	0.071162	0.083325	0.059319	0.068985	0.077266	0.060961
nqx - prob	55-59 yea	0.074529	0.093266	0.056032	0.081033	0.101557	0.061021	0.08969	0.110761	0.06935	0.092942	0.111524	0.075148	0.090554	0.104461	0.077884
nqx - prob	60-64 yea	0.109129	0.136156	0.083374	0.117247	0.14671	0.089285	0.127711	0.15846	0.098586	0.130357	0.158848	0.104724	0.128816	0.150698	0.109035
nqx - prob	65-69 yea	0.15359	0.188597	0.121216	0.163267	0.201591	0.127707	0.174734	0.215639	0.13839	0.178832	0.216341	0.145331	0.178115	0.207453	0.151354
nqx - prob	70-74 yea	0.225985	0.268161	0.188127	0.236699	0.284076	0.195592	0.252085	0.301015	0.209	0.258343	0.303026	0.218068	0.258587	0.293207	0.227034
nqx - prob	75-79 yea	0.316254	0.363439	0.277236	0.32889	0.381183	0.284388	0.34727	0.400516	0.300376	0.354165	0.403589	0.309886	0.354212	0.393428	0.32006
nqx - prob	80-84 yea	0.450025	0.498599	0.410514	0.464048	0.518952	0.417944	0.485123	0.54076	0.436775	0.4901	0.543728	0.44449	0.490555	0.534068	0.454728
nqx - prob	85+ years	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
lx - numbe	<1 year	1.00E+05	1.00E+05	1.00E+05	1.00E+05	1.00E+05	1.00E+05	1.00E+05	1.00E+05	1.00E+05	1.00E+05	1.00E+05	1.00E+05	1.00E+05	1.00E+05	1.00E+05
lx - numbe	1-4 years	96573.4	96203.29	96961.97	96163.36	95762.21	96585.67	95260.72	94756.21	95789.9	94385.21	93823.2	94978.54	93628.83	93036.99	94253.86
lx - numbe	5-9 years	95359.82	94901.48	95841.32	94586.61	94087.05	95112.55	93032.82	92417.75	93678.54	91583.37	90914.44	92289.79	90126.04	89431.75	90859.57
lx - numbe	10-14 yea	94789.13	94289.66	95313.97	93935.59	93395.84	94503.85	92267.5	91615.09	92952.69	90673.81	89970.87	91416.2	89047.92	88322.95	89813.97
lx - numbe	15-19 yea	94341.44	93771.31	94940.83	93454.47	92844.17	94097.32	91739.11	91014.11	92501.3	90089.22	89310.73	90911.92	88402.18	87608.57	89241.21
lx - numbe	20-24 yea	93747.5	93046.05	94486.11	92782.16	92037.39	93567.76	91015.5	90167.77	91908.41	89374.28	88495.04	90304.28	87715.83	86866.69	88613.65
lx - numbe	25-29 yea	92936.39	92051.56	93870.33	91873.45	90955.6	92843.68	89997.6	89023.21	91025.63	88294.84	87376.5	89266.18	86622.52	85827.59	87463.27
lx - numbe	30-34 yea	91950.16	90944.29	93013.7	90732.91	89737.18	91786.43	88640.31	87710.62	89621.12	86730.54	86032.37	87471.49	85042.31	84530.35	85588.52
lx - numbe	35-39 yea	90634.84	89530.69	91803.69	89183.66	88160.88	90265.89	86763.39	85979.74	87591.08	84557.62	84159.93	84987.59	82915.91	82687.24	83169.21
lx - numbe	40-44 yea	88818.98	87552.06	90160.76	87069.7	85957.76	88245.6	84270.86	83543.74	85039.43	81731.06	81428.56	82062.5	80257.95	80067.4	80471.67
lx - numbe	45-49 yea	86286.06	84667.65	87999.32	84199.29	82756.69	85721.73	81003.98	80020.33	82038.48	78199.11	77610.41	78821.96	76984.96	76545.78	77447.56
lx - numbe	50-54 yea	82950.62	80739.76	85290.11	80530.23	78480.44	82692.73	76961.34	75403.74	78590.18	73985.93	72872.62	75135.83	73044.32	72270.43	73838.37
lx - numbe	55-59 yea	78448.75	75325.68	81761.22	75699.23	72675.29	78893.65	71783.84	69269.23	74396.12	68720.91	66800.51	70678.89	68005.33	66686.36	69337.13
lx - numbe	60-64 yea	72602.06	68300.36	77179.99	69565.1	65294.61	74079.47	65345.52	61596.93	69236.75	62333.84	59350.64	65367.52	61847.19	59720.23	63936.86
lx - numbe	65-69 yea	64679.09	59000.87	70745.21	61408.82	55715.26	67465.3	57000.17	51836.29	62410.98	54208.17	49922.9	58521.97	53880.29	50720.5	56965.52
lx - numbe	70-74 yea	54745	47873.51	62169.78	51382.76	44483.59	58849.5	47040.31	40658.38	53773.95	44513.99	39122.53	50016.89	44283.41	40198.38	48343.59
lx - numbe	75-79 yea	42373.46	35035.7	50474	39220.53	31846.87	47339.02	35182.17	28419.6	42535.19	33014.1	27267.39	39109.82	32832.28	28411.92	37367.95
lx - numbe	80-84 yea	28972.67	22302.37	36480.8	26321.3	19707.37	33876.35	22964.44	17037.11	29758.65	21321.65	16262.56	26990.24	21202.71	17233.87	25407.95
lx - numbe	85+ years	15934.26	11182.44	21504.92	14106.95	9480.19	19717.95	11823.85	7824.114	16760.83	10871.91	7420.149	14993.35	10801.62	8029.815	13854.23
ndx - num	<1 year	3426.6	3796.708	3038.033	3836.639	4237.792	3414.332	4739.281	5243.794	4210.099	5614.789	6176.801	5021.465	6371.167	6963.014	5746.137
ndx - num	1-4 years	1213.583	1301.815	1120.645	1576.754	1675.156	1473.12	2227.903	2338.454	2111.365	2801.837	2908.756	2688.747	3502.796	3605.239	3394.294
ndx - num	5-9 years	570.6907	611.8198	527.3488	651.0165	691.213	608.6966	765.3174	802.6567	725.8456	909.5664	943.569	873.5877	1078.116	1108.793	1045.601
ndx - num	10-14 yea	447.6909	518.3429	373.1447	481.1207	551.6641	406.5356	528.3922	600.9882	451.387	584.5911	660.1422	504.2853	645.7367	714.3798	572.7603
ndx - num	15-19 yea	593.9342	725.2682	454.7193	672.3051	806.7846	529.5588	723.6098	846.3375	592.8959	714.9393	815.6951	607.6346	686.3537	741.88	627.5525
ndx - num	20-24 yea	811.1117	994.4899	615.7821	908.7191	1081.79	724.072	1017.9	1144.56	882.7736	1079.435	1118.541	1038.101	1093.312	1039.099	1150.385
ndx - num	25-29 yea	986.2318	1107.267	856.6264	1140.533	1218.42	1057.25	1357.283	1312.591	1404.514	1564.3	1344.13	1794.693	1580.206	1297.245	1874.75
ndx - num	30-34 yea	1315.317	1413.601	1210.012	1549.257	1576.304	1520.544	1876.926	1730.882	2030.04	2172.926	1872.437	2483.895	2126.399	1843.107	2419.307
ndx - num	35-39 yea	1815.862	1978.631	1642.924	2113.96	2203.113	2020.286	2492.532	2435.991	2551.646	2826.558	2731.372	2925.088	2657.965	2619.842	2697.545

ndx - num	40-44 year	2532.921	2884.405	2161.443	2870.409	3201.075	2523.871	3266.878	3523.418	3000.956	3531.953	3818.148	3240.545	3272.986	3521.622	3024.112
ndx - num	45-49 year	3335.436	3927.888	2709.208	3669.057	4276.252	3029.005	4042.637	4616.585	3448.295	4213.173	4737.791	3686.125	3940.648	4275.35	3609.188
ndx - num	50-54 year	4501.874	5414.081	3528.891	4831.002	5805.148	3799.079	5177.504	6134.509	4194.063	5265.023	6072.113	4456.945	5038.985	5584.065	4501.235
ndx - num	55-59 year	5846.689	7025.322	4581.234	6134.127	7380.674	4814.175	6438.314	7672.298	5159.371	6387.075	7449.864	5311.371	6158.142	6966.13	5400.274
ndx - num	60-64 year	7922.966	9299.495	6434.783	8156.281	9579.352	6614.173	8345.351	9760.642	6825.77	8125.663	9427.738	6845.552	7966.902	8999.735	6971.337
ndx - num	65-69 year	9934.09	11127.36	8575.424	10026.06	11231.68	8615.802	9959.862	11177.91	8637.033	9694.18	10800.37	8505.079	9596.875	10522.12	8621.935
ndx - num	70-74 year	12371.54	12837.8	11695.78	12162.23	12636.71	11510.48	11858.14	12238.78	11238.75	11499.89	11855.14	10907.06	11451.13	11786.46	10975.64
ndx - num	75-79 year	13400.79	12733.34	13993.19	12899.23	12139.5	13462.66	12217.73	11382.49	12776.54	11692.45	11004.83	12119.58	11629.57	11178.05	11960
ndx - num	80-84 year	13038.42	11119.93	14975.88	12214.36	10227.18	14158.41	11140.59	9212.995	12997.82	10449.74	8842.412	11996.88	10401.09	9204.056	11553.72
ndx - num	85+ years	15934.26	11182.44	21504.92	14106.95	9480.19	19717.95	11823.85	7824.114	16760.83	10871.91	7420.149	14993.35	10801.62	8029.815	13854.23
nLx - pers	<1 year	97601.38	97342.3	97873.38	97314.35	97033.55	97609.97	96682.5	96329.34	97052.93	96069.65	95676.24	96484.97	95540.18	95125.89	95977.7
nLx - pers	1-4 years	383381	381688.8	385158.3	380869.2	379028.5	382807.2	375695.9	373412.5	378092.3	370816.4	368311.8	373461.1	366108.6	363495.4	368869.1
nLx - pers	5-9 years	475372.4	472977.8	477888.2	471305.5	468707.2	474041	463250.8	460082.1	466578.1	455643	452213.3	459265	447934.9	444386.7	451683.8
nLx - pers	10-14 year	472826.4	470152.4	475637	468475.1	465600	471502.9	460016.5	456573	463635	451907.6	448204	455820.3	443625.3	439828.8	447637.9
nLx - pers	15-19 yea	470222.3	467043.4	473567.3	465591.6	462203.9	469162.7	456886.5	452954.7	461024.3	448658.7	444514.4	453040.5	440295	436188.2	444637.2
nLx - pers	20-24 year	466709.7	462744	470891.1	461639	457482.5	466028.6	452532.7	447977.4	457335.1	444172.8	439678.8	448926.2	435845.9	431735.7	440192.3
nLx - pers	25-29 year	462216.4	457489.6	467210.1	456515.9	451732	461575.3	446594.8	441834.6	451616.9	437563.5	433522.2	441844.2	429162.1	425894.9	432629.5
nLx - pers	30-34 year	456462.5	451187.4	462043.5	449791.4	444745.1	455130.8	438509.3	434225.9	443030.5	428220.4	425480.7	431147.7	419895.6	418044	421894.3
nLx - pers	35-39 year	448634.5	442706.9	454911.1	440633.4	435296.6	446278.7	427585.6	423808.7	431576.3	415721.7	413971.2	417625.2	407934.7	406886.6	409102.2
nLx - pers	40-44 year	437762.6	430549.3	445400.2	428172.5	421786.1	434918.3	413187.1	408910.2	417694.8	399825.4	397597.4	402211.2	393107.3	391533	394798.1
nLx - pers	45-49 year	423091.7	413518.5	433223.6	411823.8	403092.8	421036.2	394913.3	388560.2	401571.7	380462.6	376207.6	384894.5	375073.2	372040.5	378214.8
nLx - pers	50-54 year	403498.4	390163.6	417628.3	390573.6	377889.3	403965.9	371862.9	361682.4	382465.8	356767.1	349182.8	364536.8	352624.1	347392	357938.7
nLx - pers	55-59 year	377627	359065.1	397353	363160.8	344924.8	382432.8	342823.4	327165.4	359082.2	327636.9	315377.9	340116	324631.3	316016.5	333185
nLx - pers	60-64 year	343202.9	318253.1	369813	327434.8	302524.7	353861.9	305864.2	283583.1	329119.3	291355	273183.9	309723.7	289318.7	276101.8	302255.9
nLx - pers	65-69 year	298560.2	267185.9	332287.5	281979	250497.1	315787	260101.2	231236.7	290462.3	246805.4	222613.6	271347.1	245409.2	227297.2	263272.8
nLx - pers	70-74 year	242796.2	207273	281609.4	226508.2	190826.1	265471.3	205556.2	172694.9	240772.9	193820.2	165974.8	222816.8	192789.2	171525.8	214278.8
nLx - pers	75-79 year	178365.3	143345.2	217387	163854.6	128885.6	203038.4	145366.5	113641.8	180734.6	135839.4	108824.9	165250.2	135087.5	114114.5	156939.7
nLx - pers	80-84 year	112267.3	83712	144964.3	101070.6	72968.91	133985.7	86970.75	62153.06	116298.7	80483.88	59206.78	104959	80010.81	63159.21	98155.45
nLx - pers	85+ years	77142.97	49604.15	111542.6	67827.08	41329.86	102504.3	55348.96	33167.23	84760.53	50613.29	31370.48	75113.33	50026.31	34212.24	68246.84
Tx - perso	<1 year	6627741	6366003	6916389	6454541	6196555	6741139	6199749	5969993	6452904	6012383	5821113	6218584	5924420	5774979	6079910
Tx - perso	1-4 years	6530140	6268660	6818516	6357226	6099521	6643529	6103067	5873664	6355851	5916313	5725437	6122099	5828880	5679853	5983933
Tx - perso	5-9 years	6146759	5886971	6433357	5976357	5720493	6260722	5727371	5500251	5977759	5545497	5357125	5748638	5462771	5316358	5615063
Tx - perso	10-14 year	5671387	5413994	5955469	5505051	5251785	5786681	5264120	5040169	5511181	5089854	4904911	5289373	5014836	4871971	5163380
Tx - perso	15-19 yea	5198560	4943841	5479832	5036576	4786185	5315178	4804104	4583596	5047546	4637946	4456707	4833552	4571211	4432142	4715742
Tx - perso	20-24 year	4728338	4476798	5006265	4570985	4323981	4846015	4347217	4130642	4586521	4189288	4012193	4380512	4130916	3995954	4271104
Tx - perso	25-29 year	4261628	4014054	4535374	4109346	3866499	4379987	3894684	3682664	4129186	3745115	3572514	3931586	3695070	3564218	3830912
Tx - perso	30-34 year	3799412	3556564	4068164	3652830	3414767	3918412	3448090	3240830	3677569	3307551	3138992	3489741	3265908	3138323	3398283
Tx - perso	35-39 year	3342949	3105377	3606120	3203038	2970022	3463281	3009580	2806604	3234539	2879331	2713511	3058594	2846012	2720279	2976388
Tx - perso	40-44 year	2894315	2662670	3151209	2762405	2534725	3017002	2581995	2382795	2802963	2463609	2299540	2640969	2438078	2313393	2567286
Tx - perso	45-49 year	2456552	2232121	2705809	2334233	2112939	2582084	2168808	1973885	2385268	2063784	1901943	2238757	2044970	1921860	2172488
Tx - perso	50-54 year	2033460	1818602	2272585	1922409	1709846	2161048	1773894	1585325	1983696	1683321	1525735	1853863	1669897	1549819	1794273
Tx - perso	55-59 year	1629962	1428438	1854957	1531835	1331957	1757082	1402031	1223642	1601230	1326554	1176552	1489326	1317273	1202427	1436335
Tx - perso	60-64 year	1252335	1069373	1457604	1168674	987032.3	1374649	1059208	896476.7	1242148	998917.2	861174.4	1149210	992641.8	886410.7	1103150
Tx - perso	65-69 year	909132	751120.3	1087791	841239.5	684507.6	1020787	753343.7	612893.7	913029	707562.2	587990.5	839486.4	703323.1	610308.9	800893.6
Tx - perso	70-74 year	610571.8	483934.4	755503.4	559260.5	434010.5	704999.8	493242.5	381657	622566.7	460756.8	365376.9	568139.2	457913.8	383011.7	537620.9
Tx - perso	75-79 year	367775.6	276661.3	473894	332752.3	243184.4	439528.5	287686.3	208962.1	381793.8	266936.5	199402.1	345322.5	265124.6	211485.9	323342
Tx - perso	80-84 year	189410.3	133316.2	256507	168897.7	114298.8	236490.1	142319.7	95320.29	201059.2	131097.2	90577.25	180072.3	130037.1	97371.46	166402.3
Tx - perso	85+ years	77142.97	49604.15	111542.6	67827.08	41329.86	102504.3	55348.96	33167.23	84760.53	50613.29	31370.48	75113.33	50026.31	34212.24	68246.84
ex - expec	<1 year	66.27741	63.66003	69.16389	64.54541	61.96555	67.41139	61.99749	59.69993	64.52904	60.12383	58.21113	62.18584	59.2442	57.74979	60.7991
ex - expec	1-4 years	67.61841	65.16056	70.32155	66.10861	63.69445	68.7838	64.06698	61.98712	66.352	62.68263	61.02368	64.45771	62.25518	61.04941	63.4874
ex - expec	5-9 years	64.45858	62.03245	67.12509	63.18397	60.8	65.82435	61.56291	59.5151	63.8114	60.55135	58.9249	62.28899	60.61257	59.44598	61.79936
ex - expec	10-14 year	59.83161	57.41874	62.48265	58.60453	56.23147	61.23222	57.05281	55.01462	59.29017	56.13367	54.51666	57.86034	56.31615	55.16087	57.48972
ex - expec	15-19 yea	55.10368	52.72232	57.71839	53.89337	51.55073	56.48597	52.36702	50.36138	54.56729	51.4817	49.90114	53.16742	51.70925	50.59028	52.84265
ex - expec	20-24 year	50.43695	48.11379	52.98414	49.26577	46.98071	51.79151	47.76348	45.81062	49.90318	46.87353	45.33806	48.50835	47.0943	46.00099	48.19917
ex - expec	25-29 year	45.85532	43.60658	48.31531	44.72833	42.50974	47.17593	43.27543	41.36746	45.3629	42.41601	40.88644	44.0434	42.65715	41.52765	43.80024
ex - expec	30-34 year	41.32034	39.10706	43.73725	40.25915	38.05298	42.69053	38.89979	36.94911	41.03463	38.13595	36.48618	39.89576	38.40333	37.12659	39.70489
ex - expec	35-39 year	36.88371	34.68505	39.28078	35.91508	33.68866	38.36755	34.68721	32.64262	36.92772	34.0517	32.24232	35.98871	34.32408	32.89842	35.78714
ex - expec	40-44 year	32.58667	30.41242	34.951	31.72637	29.48803	34.18869	30.63924	28.52152	32.96074	30.14288	28.23997	32.1824	30.37802	28.89307	31.90298
ex - expec	45-49 year	28.46986	26.36332	30.74806	27.72271	25.53195	30.12169	26.77409	24.66729	29.07499	26.3914	24.50628	28.40271	26.56324	25.10732	28.05109
ex - expec	50-54 year	24.51411	22.52424	26.64535	23.87189	21.78691	26.13346	23.04916	21.02448	25.24102	22.75191	20.93701	24.67349	22.86143	21.44472	24.30001
ex - expec	55-59 year	20.77741	18.9635	22.68749	20.23581	18.32751	22.27152	19.5313	17.66502	21.52304	19.3035	17.61292	21.07172	19.37014	18.03108	20.71523
ex - expec	60-64 year	17.2493	15.65692	18.88577	16.79972	15.1166	18.55641	16.20934	14.55392	17.94059	16.02528	14.50994	17.58075	16.04991	14.84272	17.25373
ex - expec	65-69 year	14.05604	12.73067	15.37618	13.699	12.28582	15.13055	13.21652	11.82364	14.6293	13.05268	11.77797	14.34481	13.05344	12.03279	14.05927
ex - expec	70-74 year	11.15301	10.1086	12.15226	10.88421	9.756644	11.97971	10.48553	9.386921	11.57748	10.35083	9.339297	11.35895	10.34053	9.528038	11.12083
ex - expec	75-79 year	8.679386	7.896554	9.388873	8.484135	7.636052	9.284699	8.177046	7.352745	8.975951	8.085531	7.312843	8.829558	8.07512	7.443563	8.652925
ex - expec	80-84 year	6.53755	5.977669	7.031286	6.416768	5.799797	6.980979	6.197394	5.594863	6.756329	6.148548	5.569679	6.671757	6.133043	5.650005	6.549222
ex - expec	85+ years	4.841329	4.435899	5.186842	4.808063	4.359602	5.19853	4.681127	4.239104	5.057062	4.65542	4.227742	5.009775	4.631373	4.260652	4.926065

4. DATA DESCRIPTION

Data was obtained from Ghana Statistical Service. The year 2015 was selected together with some of the actuarial values needed for our computations. A radix of 10,000 was used and an omega of 90 was used in our life table.

Age(x)	l_x	new l_x	${}_nd_x$	${}_nq_x$	${}_np_x$	S_x	e_x	A_x	$\ddot{a}$	P_x
0	100,000.00	10,000.00	383.66	0.038366	0.961634	1	64.54541	0.06136	16.5825545	0.003701
5	96163.361	9,616.34	157.68	0.016397	0.983603	0.961634	66.10861	0.04550	16.8628467	0.002698
10	94586.607	9,458.66	65.10	0.006883	0.993117	0.945866	63.18397	0.04523	16.867545	0.002682
15	93935.59	9,393.56	48.11	0.005122	0.994878	0.939356	58.60453	0.05402	16.7122882	0.003232
20	93454.47	9,345.45	67.23	0.007194	0.992806	0.934545	53.89337	0.06752	16.4738691	0.004098
25	92782.165	9,278.22	90.87	0.009794	0.990206	0.927822	49.26577	0.08376	16.1868819	0.005175
30	91873.445	9,187.34	114.05	0.012414	0.987586	0.918734	44.72833	0.10331	15.8415347	0.006521
35	90732.912	9,073.29	154.93	0.017075	0.982925	0.907329	40.25915	0.12742	15.4156004	0.008266
40	89183.655	8,918.37	211.40	0.023703	0.976297	0.891837	35.91508	0.15611	14.9087987	0.010471
45	87069.695	8,706.97	287.04	0.032967	0.967033	0.870697	31.72637	0.18970	14.3153403	0.013251
50	84199.287	8,419.93	366.91	0.043576	0.956424	0.841993	27.72271	0.22842	13.6312134	0.016757
55	80530.23	8,053.02	483.10	0.05999	0.94001	0.805302	23.87189	0.27405	12.8251885	0.021368
60	75699.228	7,569.92	613.41	0.081033	0.918967	0.756992	20.23581	0.32632	11.9016562	0.027418
65	69565.101	6,956.51	815.63	0.117247	0.882753	0.695651	16.79972	0.38702	10.8293128	0.035738
70	61408.82	6,140.88	1,002.61	0.163267	0.836733	0.614088	13.699	0.45389	9.64793353	0.047045
75	51382.764	5,138.28	1,216.22	0.236699	0.763301	0.513828	10.88421	0.53080	8.28913884	0.064036
80	39220.534	3,922.05	1,289.92	0.32889	0.67111	0.392205	8.484135	0.62051	6.70432318	0.092554
85	26321.304	2,632.13	2,632.13	1	0	0.263213	6.416768	0.74726	4.46510558	0.167355
90	0	0.00	0.00	0	0	0	0	0.00000	0	0

Computations made such as the finding the discount and the discounting factor are shown below.

$v = (1+i)^{-1} = (1+6\%)^{-1} = 0.9433962264$				
		v	0.943396226	
		v^5	0.747258173	
		d=1-v		
		d	0.056603774	

5. COMPARING SURVIVAL CURVES AND MAPE

The Mean Absolute Percentage Error is computed to find the error between the survival curves.
To find our Mean Absolute Percentage Errors we have to make use of the formula in the preamble which states that;

$$\text{MAPE} = \frac{100}{\text{N}} * \sum_{t=1}^{N} \left| \frac{Y_t - T_t}{Y_t} \right|$$

The MAPE which is the least gives us a better prediction.

Using the predicted alpha values(α), we had our table giving us;

Age(x)	l_x	new l_x	$_nd_x$	$_nq_x$	$_np_x$	S_x	e_x	Ut-S(x)	Ut-S(x)	Ut-S(x)	Ut-S(x)	Ut-S(x)
0	100,000.00	10,000.00	383.66	0.038366	0.961634	1	64.54540547	0	0	0	0	0
5	96163.36089	9,616.34	157.68	0.016397	0.983603	0.961634	66.10861076	0.015762	0.012973	0.010192	0.007418	0.004653
10	94586.60679	9,458.66	65.10	0.006883	0.993117	0.945866	63.18396615	0.008113	0.002512	-0.00306	-0.00859	-0.0141
15	93935.59029	9,393.56	48.11	0.005122	0.994878	0.939356	58.60453372	-0.00969	-0.01813	-0.02648	-0.03477	-0.04298
20	93454.46964	9,345.45	67.23	0.007194	0.992806	0.934545	53.89337009	-0.03018	-0.04148	-0.05263	-0.06364	-0.07452
25	92782.16452	9,278.22	90.87	0.009794	0.990206	0.927822	49.26576953	-0.04987	-0.06404	-0.07799	-0.0917	-0.1052
30	91873.44539	9,187.34	114.05	0.012414	0.987586	0.918734	44.72832918	-0.06845	-0.08552	-0.10224	-0.11862	-0.13468
35	90732.91199	9,073.29	154.93	0.017075	0.982925	0.907329	40.25914891	-0.08613	-0.1061	-0.12559	-0.14461	-0.16316
40	89183.65539	8,918.37	211.40	0.023703	0.976297	0.891837	35.91508312	-0.10136	-0.12425	-0.14648	-0.16807	-0.18903
45	87069.69513	8,706.97	287.04	0.032967	0.967033	0.870697	31.72636606	-0.11284	-0.13865	-0.16359	-0.18768	-0.21094
50	84199.28662	8,419.93	366.91	0.043576	0.956424	0.841993	27.72271191	-0.11901	-0.14774	-0.17533	-0.20182	-0.22725
55	80530.23003	8,053.02	483.10	0.05999	0.94001	0.805302	23.87188977	-0.11992	-0.15154	-0.18169	-0.21046	-0.23789
60	75699.22828	7,569.92	613.41	0.081033	0.918967	0.756992	20.23580905	-0.1126	-0.14704	-0.17964	-0.2105	-0.23971
65	69565.10106	6,956.51	815.63	0.117247	0.882753	0.695651	16.79972123	-0.09658	-0.13374	-0.1686	-0.2013	-0.23197
70	61408.82031	6,140.88	1,002.61	0.163267	0.836733	0.614088	13.69900118	-0.06617	-0.10587	-0.14268	-0.17683	-0.20851
75	51382.76358	5,138.28	1,216.22	0.236699	0.763301	0.513828	10.88420519	-0.02547	-0.06732	-0.10558	-0.14056	-0.17255
80	39220.53421	3,922.05	1,289.92	0.32889	0.67111	0.392205	8.484134756	0.023038	-0.02016	-0.05887	-0.09355	-0.12462
85	26321.30379	2,632.13	2,632.13	1	0	0.263213	6.416768097	0.051483	0.009137	-0.02751	-0.05923	-0.08668
90	0	0.00	0.00	0	0	0	0	0	0	0	0	0

Using the MAPE formula, we get our MAPE table as shown below;

	finding the MAPE's for the different alpha values				
	0.4	0.45	0.5	0.55	0.6
	0	0	0	0	0
	0.01639129	0.01349067	0.01059833	0.00771424	0.00483838
	0.00857766	0.00265545	0.00323199	0.00908485	0.01490335
	0.01031421	0.01929527	0.02819482	0.03701362	0.04575238
	0.03229689	0.0443807	0.05631362	0.06809753	0.07973429
	0.05375447	0.06902637	0.08405178	0.09883469	0.11337901
	0.07450624	0.0930801	0.1112812	0.12911702	0.14659489
	0.09492781	0.116942	0.13842073	0.15937704	0.17982362
	0.11364902	0.13931922	0.16424597	0.1884508	0.21195462
	0.1295958	0.15924496	0.18788417	0.21554781	0.24226913
	0.14134524	0.17546432	0.20822765	0.23968912	0.26990046
	0.14891826	0.18817476	0.22562054	0.26133911	0.29541015
	0.14874427	0.19424319	0.23731023	0.27807537	0.31666165
	0.1388326	0.19225839	0.24236971	0.28937218	0.33345868
	0.10775476	0.17239403	0.23235047	0.28796333	0.33954727
	0.04956583	0.1310106	0.20547619	0.27356066	0.33581083
	0.05874041	0.05141257	0.15010507	0.23852944	0.31775401
	0.19559365	0.03471317	0.10451905	0.22501602	0.32929878
	1	1	1	1	1
TOTAL	2.52350843	2.79710576	3.39020151	4.00678284	4.5770915
MAPE's:	13.2816233	14.7216093	17.8431659	21.0883307	24.0899553

Below is a summarized table of alpha values(α) and their corresponding MAPE's

α values	0.4	0.45	0.5	0.55	0.6
MAPE	13.2816%	14.7216%	17.8432%	21.0883%	24.0900%

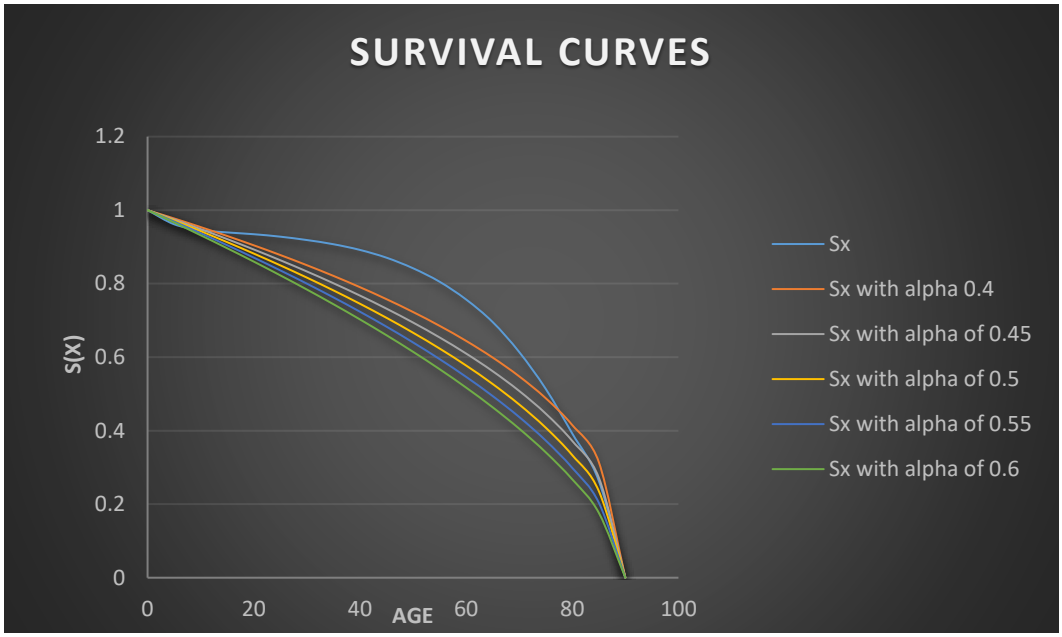
Since an alpha value(α) of 0.4 yields the least Mean Absolute Percentage Error, we conclude that an alpha(α) of 0.4 is our preferred choice.

6. SURVIVAL CURVES

The survival curves is derived by plotting the survival points against the age. We first plot the Survival points making up the least MAPE together with the original survival points against age and then we do same for our second graph. The only difference in the second graph is that we gather all possible survival points and plot them against age to get a new graph.

S_x with the new alpha values				
0.4	0.45	0.5	0.55	0.6
1	1	1	1	1
0.977396	0.974606688	0.971825	0.969052	0.966286
0.953979	0.948377769	0.942809	0.937273	0.931769
0.929667	0.921230778	0.912871	0.904587	0.896378
0.904362	0.893068947	0.881917	0.870905	0.860029
0.877947	0.863777488	0.849837	0.836121	0.822626
0.850283	0.833218556	0.816497	0.80011	0.784053
0.821198	0.801224242	0.781736	0.762722	0.74417
0.79048	0.767586579	0.745356	0.723769	0.702808
0.757858	0.732042848	0.707107	0.68302	0.659754
0.722981	0.694253163	0.666667	0.640176	0.614739
0.685378	0.653764735	0.62361	0.594845	0.567408
0.644394	0.609951685	0.57735	0.546491	0.517282
0.599072	0.561906265	0.527046	0.494349	0.46368
0.547917	0.508223062	0.471405	0.437253	0.405576
0.488359	0.446510769	0.408248	0.373265	0.341279
0.415244	0.372041058	0.333333	0.298653	0.267581
0.314696	0.272349996	0.235702	0.203986	0.176537
0	0	0	0	0





7. CONCLUSION

We were asked to generate a life table with a radix of 10,000 using the life table from World Health Organization as a sample.

The general Demoivre distribution $S(x) = \left(\frac{\omega - x}{\omega}\right)^{\alpha}$ was used using alphas of 0.4, 0.45, 0.5, 0.55, and 0.60. From our calculations $\alpha = 0.4$ was chosen since it gave us the least absolute margin of error.